

APPLICATIONS OF ĆIRIĆ CONTRACTIONS

1. NUMERICAL ANALYSIS & OPTIMIZATION

- Fixed point iteration methods
- Root finding algorithms
- Convergence analysis of iterative schemes

2. DIFFERENTIAL & INTEGRAL EQUATIONS

- Existence and uniqueness of solutions
- Solution of nonlinear ODEs
- Integral equations (Hammerstein, Volterra type)

3. FUNCTIONAL ANALYSIS

- Approximation theory
- Best approximation problems
- Iterative approximation methods

4. CONTROL THEORY & SYSTEMS

- Stability analysis
- Lyapunov functions
- Feedback system design

5. NONLINEAR ANALYSIS

- Variational problems
- Nonlinear operator equations
- Monotone operators on Banach spaces

6. COMPUTATIONAL MATHEMATICS

- Machine learning algorithms (gradient descent)
- Optimization algorithms
- Convergence guarantees